

Zero-Harm Regulatory Compliance Mapping

Zero-Harm — Regulatory Compliance Mapping

Why this file matters

The official brief asks for “**regulatory compliance coverage (OISD/Factory Act/DGMS)**” as a named evaluation focus. Most teams will skip this because it takes extra research. Doing it — even simply — is an easy way to look more serious than other teams, with very little extra coding.

This table shows exactly which law/standard each part of your system already satisfies, using words a non-technical judge can follow in 10 seconds.

Mapping Table

Zero-Harm Feature	What it does	Regulation it satisfies	In simple words
Permit Agent (hot-work / confined-space tracking)	Records every active permit, its type, zone, and validity window	Factory Act 1948, Section 36 & 38 (precautions against dangerous fumes, and work in confined spaces)	The law says: before risky work starts, someone must check it's safe. Our system checks this automatically, all the time — not just once at sign-off.
Gas Agent (threshold monitoring)	Tracks gas levels per zone against WARNING/CRITICAL thresholds	OISD-STD-105 (process safety management for oil & gas / hazardous plants) and OISD-STD-106 (fire/gas detection systems)	OISD says plants must have working gas detection and act on it. Our system doesn't just detect — it also <i>decides</i> what action to take.
Compound Risk Alert (permit + gas + maintenance fusion)	Flags dangerous combinations before they escalate	Factory Act 1948, Section 7A (general duty of occupier to ensure health & safety)	The Act places responsibility on the plant owner to <i>prevent</i> harm, not just record it after. Our alerts exist to prevent, in real time.
Maintenance Agent (CMMS activity tracking)	Flags maintenance jobs running during elevated-risk	DGMS Circular on Permit-to-Work Systems (Directorate	DGMS requires maintenance and permit systems to be

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	conditions	General of Mines Safety)	cross-checked. Ours does this automatically instead of relying on a supervisor remembering to check two separate paper systems.
Alert log with triggers + recommended action	Every alert stores what caused it and what to do next	Factory Act 1948, Section 87 / Rules on accident reporting	Regulations require plants to document safety incidents and near-misses with evidence. Our alert log is already an audit trail — this is a natural evidence source if an inspector asks “how did you know, and what did you do?”

What to say to a judge (one sentence per row)

“Our system isn’t just detecting danger — every single feature we built maps directly to an existing Indian safety law or standard. That means this isn’t just a smart alarm, it’s a system that helps a plant *prove* it followed the law, automatically.”

Honest roadmap note (say this too — it builds trust)

We have **not** built a full RAG-based compliance checker that reads full OISD/DGMS PDF documents and cross-references clause-by-clause — that is listed as our Section 13 roadmap item. What we *have* done is map our existing implemented data fields (zone, permit type, timestamps, gas readings) to the specific clauses above, so the extension is a data-plumbing exercise, not a redesign.

Being honest about what’s built vs. what’s roadmap is itself a credibility signal — judges are experienced enough to notice teams who overclaim.